

Calophyllum Oil Prospective as Alternate Fuel for Diesel Engine

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Abstract: Rapid depletion of fossil fuels, increasing fossil-fuel price, carbon price, and the quest of low carbon fuel for cleaner environment – these are the reason researchers are looking for alternatives of fossil fuels. Biodiesel is a gifted substitute as an alternative fuel has gained significant attention due to the predicted littleness of conventional fuels and environmental concern. The utilization of liquid fuels such as biodiesel produced from Calophyllum inophyllum oil by transesterification process represents one of the most promising options for the use of conventional fossil fuels. The Calophyllum inophyllum oil is converted into Calophyllum inophyllum methyl ester known as biodiesel processed in the presence of homogeneous acid catalyst. The physical properties such as Kinematic viscosity, Density, Calorific Value, Cetane number, Fire point and Flash point were found out for Calophyllum inophyllum methyl ester at different blends.

Keywords: Calophyllum inophyllum, transesterification, Biodiesel, Alternative fuel, Emission

I. INTRODUCTION

In 2040, the demand for global energy will be about 30% higher than that of in 2010. In 2020, the global demand for crude oil is estimated to be 101.6 million barrels per day. The fossil fuels account for about 86% of the primary energy need which is increasingly rapidly. Most industries depend upon diesel as fuel for their energy needs. Mankind relies heavily on fuels for every aspect of their everyday life, good transport and electricity in particular. Since the last few decades, there is a drastic increase in demand of fuel, price and no. of a road vehicle, climate changes due to air pollutions and now it has become a serious topic for the researchers to guarantee the availability of energy alternatives and reduction in environmental pollution due to vehicular emissions.

1.1 Need of Biodiesel

Biodiesels are gaining worldwide acceptance because of the faster depleting petro-diesel fuels. Biodiesels are biodegradable, renewable and more environment friendly than petroleum based fuels [1].

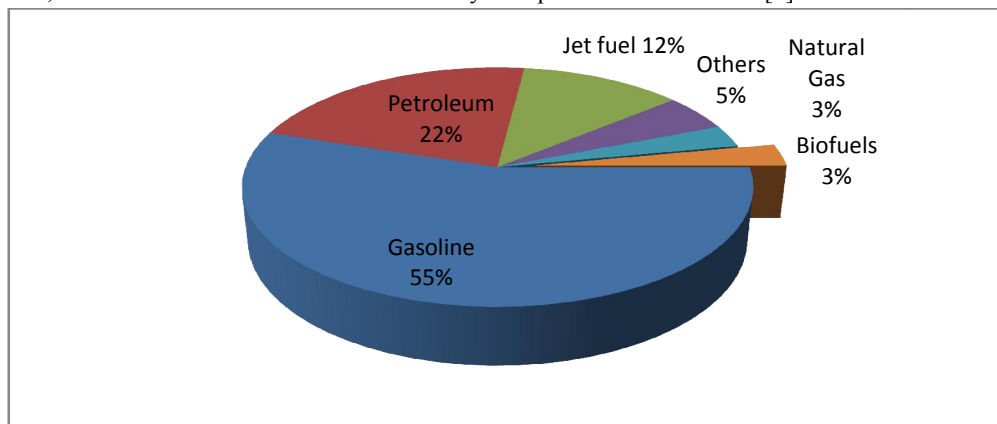


Figure 1: Transportation Energy Consumption

Ever increasing petroleum products prices and uncertainties concerning their availability have Increased the importance of vegetable oil based biofuel tremendously. The biodiesel have some rewards as compared to petroleum diesel [2-3]. The worldwide energy consumption rate is currently growing rapidly than the population growth rate, and is becoming a critical global issue. It has been reported that over the last 35 years, global primary fuel consumption has doubled [4]. The most important advantages of biodiesel are higher flash point, biodegradability, improved cetane number and reduced exhaust. From [Fig. 1] it can be seen that fossil fuels hold a huge share in the transportation energy consumption, and merely only 3% accounts for energy consumption. Fig. 2, shows us the Biodiesel market share in different Continents where Asia Lies 5% only.

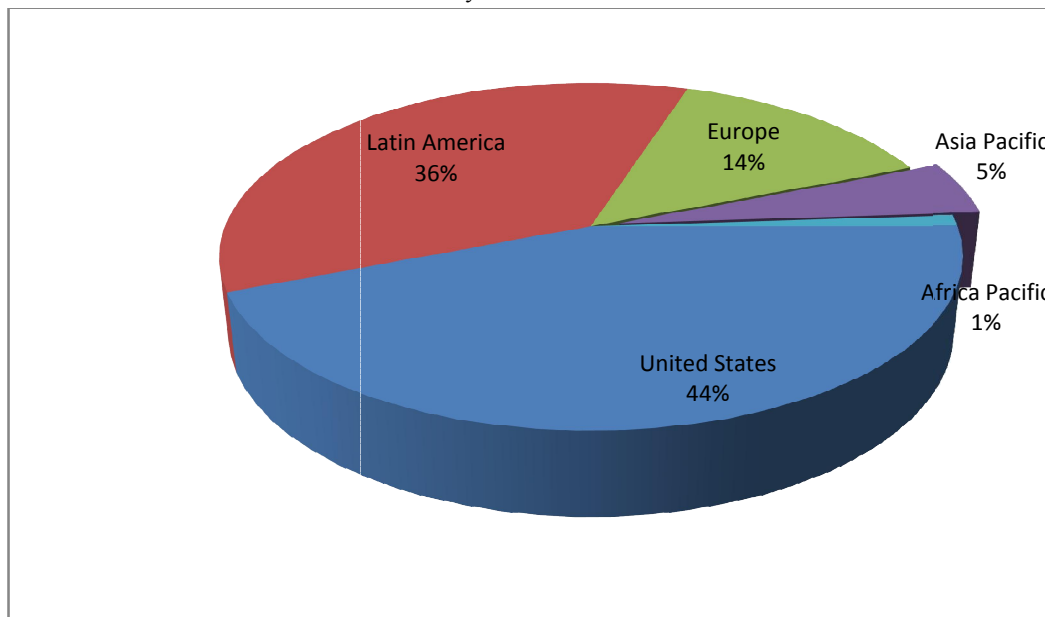


Figure 2: World Market Share of Biodiesel

1.2 Calophyllum Oil Over Other Oils

Some of the non-edible oil widely used for the process of biodiesel production are *Jatropacurcas*, *Calophyllum inophyllum*, *Pongaminapinnata*, *Ricinus Communis* and *Ceiba Pentantra*. *Calophyllum inophyllum* fruit and seeds shown in Fig. 1 are considered as a suitable feedstock for the production of biodiesel. *Calophyllum inophyllum* is a species of family Guttifereae (Clusiaceae), it is a broad leaved evergreen tree widely planted throughout the tropics and easily grown along the coastal area. Some of its advantages over *Jatrophais* that the *Calophyllum* tree has a high yield of oil of 4560 kg per hectare as compared to oil yield of 1560 per hectare for *Jatropha* [1].

One of the major promising processes to transform from vegetable oil to methyl ester is the transesterification, in which alcohol reacts with triglycerides of free fatty acids (vegetable oil) in the presence of catalyst like NaOH/KOH. *Jatropha* vegetable oil is one of the leading non edible sources existing in India. The *Calophyllum* seed has a very high oil content of 65–75% but *Jatropha* seed has an oil content of only 55–60%. One yet major limitation of *Calophyllum inophyllum* feedstock is that its rich content of free fatty acids, which should be properly removed during transesterification process [2,4].

Calophyllum Oil is chosen for the investigation of diesel engine performance and exhaust emission and optimisation. The motivation behind the selection of *Calophyllum* oil for this study is, it posses the position of second highest oil yield [Fig. 1] of 4680 lit/hectare after palm after Palm which reported as 5950 lit./hectare in a category of non-edible oil feedstock [4].

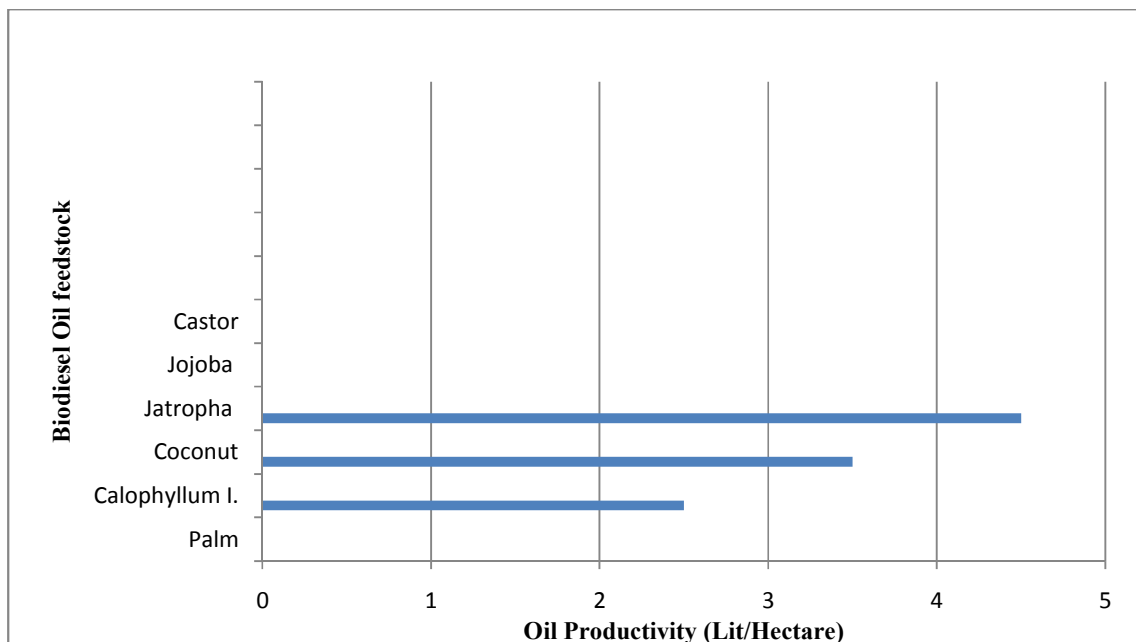


Table 1: Biodiesel feedstock oil productivity yield.

1.3 Production of Biodiesel

When Rudolf Diesel used biodiesel on engines it did not burn completely due to high viscosity and volatility of vegetable oils and it forms deposits on fuel injectors. Later biodiesel which meets the requirements of ASTM stands is produced which has fuel properties similar to that of petroleum diesel [3]. Calophyllum inophyllum seed is one of the easily cultivable and widely available feedstock for biodiesel transesterification in developing countries like India. Only very few research works have been carried out in implementing 100% Calophyllum inophyllum methyl ester and blends as alternative fuel for diesel. [4,5]. Calophyllum inophyllum fruit in tree and dried seed shown in [Fig. 3].



Figure 3: Calophyllum inophyllum fruit in tree and dried seeds.

In a short period of time, from 2001 to 2010, biodiesel production increased to 294,690 barrels per day, whereas, consumption increased to 313,770 barrels per day from 16,490 barrels per day [7]. Advantage of biodiesel is that it is environment friendly, renewable, non-flammable, biodegradable, and non-toxic. There are many ways and procedures to biodiesel fuel from vegetable oil such as pyrolysis, dilution, microemulsion and transesterification Biodiesel produced via transesterification process has proven to be a viable alternative fuel with similar characteristics to diesel

fuel. Besides, some researchers are using two-step esterification and transesterification processes in producing biodiesel from crude oils with high free fatty acid (FFA) [8].

Unfortunately, Calophyllum inophyllum oil usually contains large amount of FFA (~22 wt.%) and therefore cannot be converted into biodiesel following the conventional alkali-catalyzed transesterification reaction. A two or three stage method has been applied to produce high quality biodiesel from this feedstock by few researchers [9-11]. Olubunmi O et al. [9] utilized a three-stage transesterification process to produce biodiesel from C. inophyllum oil with initial acid value of 44 mg KOH/g. Also, biodiesel was produced from the oil of C. inophyllum via acid pre-treatment, transesterification and post treatment stages.

A typical procedure for the extraction of Calophyllum oil: The seeds were dried and the kernels were finely powdered and the powder was soxhlet extracted using hexane as a solvent. The hexane was removed under vacuum and the crude oil (40%) was dried under reduced pressure. The crude oil is used without further purification. Complete production process of Calophyllum process is shown below in Fig. 4

- **Esterification with Acid Catalyst to Reduce Ester:** In this stage FFA (free fatty acid) is esterified with methanol and acid catalyst to produce ester. Molar ratio of alcohol to oil is one of the main parameters which affects the conversion efficiency, biodiesel yield and production. The optimum conditions for the reactants were found to be 16:1 methanol to oil ratio, 10 ml of H₂SO₄ for 100 ml of oil, reaction temperature of 60°C, and 45 min of reaction time along with a constant stirring rate on a magnetic stirrer.
- **Transesterification with Alkali Catalyst:** Since FFA was converted to ester in the first step, we use alkali catalyst for transesterification process. The optimum conditions for the reactants were found to be 16:1 methanol to oil ratio, 0.5 g concentration of Sodium hydroxide for 100 ml of oil, reaction temperature of 60°C, 30 min of reaction time along with a constant stirring rate.
- **Purification:** Purification is done by removing methanol while heating biodiesel at 75°C for about 30 min to remove trapped methanol and the yield of biodiesel was found to be 85%. The various physical and chemical properties of CIME have been evaluated and compared with diesel.

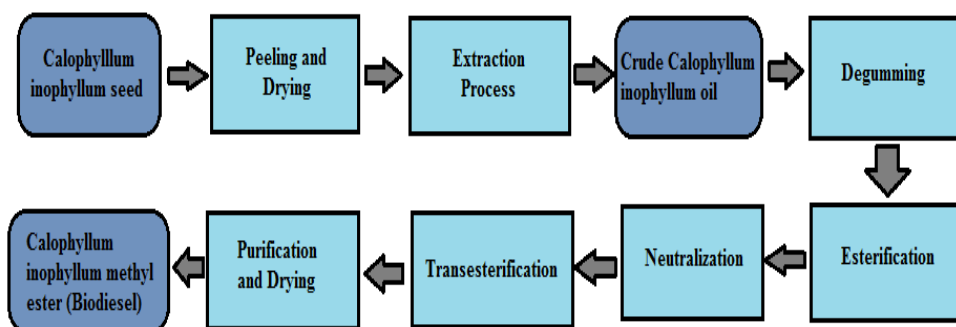


Figure 4: Flow chart methodology of Calophyllum inophyllum biodiesel production process.

1.4 Thermophysical Properties of Calophyllum Oil

The specific gravity reduces after transesterification, viscosity from 38.17 to 3.7 centistokes, which is acceptable as per ASTM norms for Biodiesel. Flash point and fire point are important temperatures specified for safety during storage, handling and transport. The flash point and fire point of biodiesel was found to be 24°C and 34°C respectively by PMCC method. Flash point of Calophyllum inophyllum oil decreases after transesterification, which shows that its volatile characteristics had developed and it is also safe to handle. T.Elangovan et al. [2] studied different properties of diesel with various blends like B20, B40, B60, B80, B100 etc. and is shown in the table below Table.1. The blend shows better atomization of fuel droplets, promote better mixing of air-fuel and leads to enhance combustion.

Fuel Properties	Unit	B20	B40	B60	B80	B100	DIESEL
Density @15 ⁰ C	Kg/m ³	846.6	854.6	857.5	867.3	868.6	840
Kinematic viscosity @40 ⁰ C	Cst	2.85	3.03	3.3	3.54	3.7	3.12
Flash Point	⁰ C	30	28	36	36	24	25
Fire Point	⁰ C	40	38	36	36	34	36
Calorific Value	MJ	43.86	41.21	40.13	39.02	36.86	44.34
Cetane Number		56	55	54	53	52	48

Table 1: Calophyllum inophyllum Biodiesel Properties at different Blends

1.5 Performance and Emissions of Calophyllum Oil

From the available literature, It was witnessed that the current engines are designed for only the diesel fuel, therefore, the engine for biodiesel is comparatively less in performance. This may also due to the less calorific value of biodiesel. However, some researcher have reported that by increasing power output upto B20 as compared to diesel, as the biodiesel inherently having oxygen content approximately 10% in weight that improve combustion specially at rich zone which result in more complete combustion [16].

Comparison for engine output performance of fuels with different calorific values, BSEC is more reasonable than BSFC as it finds out the amount of energy need to be spent to produce unit power. Brake specific energy consumption is the ratio of energy obtained by burning of fuel to the actual energy, i.e. brake power. It indicates how effectively energy obtained from the combustion of the fuel. [Fig. 5], represents the variation in BSEC and it showing decreasing trend for all of the fuel blend and neat diesel with the increase in compression ratio as 14:1, 15:1, 16:1 and 17:1 and engine load conditions as 25%,50%75%and 100% of the engine full load conditions [17].

Also as engine load is increased, the overall engine temperature and cycle pressure increase which favours the better combustion characteristics and higher output. Below graph shows us the variations in Brake Specific Energy Consumption for different Blends at Various Loads at Compression ratio of 17. [16].

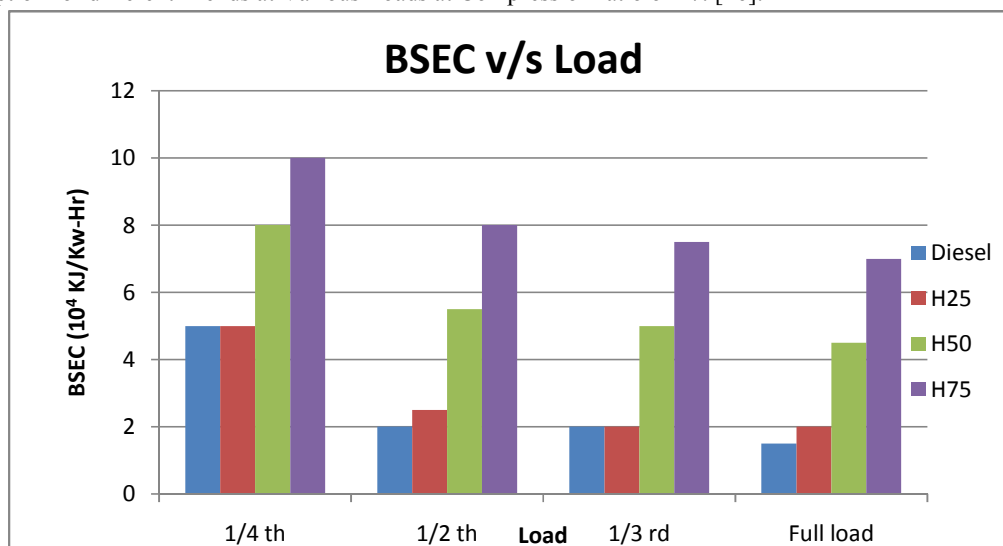


Table 2: Brake Specific energy consumption characteristics over different loads

The study found that there is increase in brake thermal efficiency (BTE) upto 8.9%, CO emission increased upto 50%, smoke emission upto 35% but at the same time it is reported that there is decrease in HC emission upto 22%, NOx emission 6.8%.

Exhaust gas temperature (EGT) changes according to engine speed and load conditions, high load and high-speed condition leads to higher exhaust gas temperature emissions. The exhaust gas temperature increases with engine load, as at the higher load condition engine needed to developed higher output which also results in higher exhaust gas temperature.

Carbon monoxide (CO) emission is an indication of deficiency of oxygen which causes incomplete combustion, thus it is susceptible to the fuel-rich zones, poor mixing. At part load condition fuel injection is lesser and the ample amount of oxygen available which causes lesser CO emission and at full load conditions the amount of oxygen available is lesser for the increased rate of fuel injection thus this increase the CO emission even though engine gives better and higher output.

Hydrocarbon (HC) emissions are first and foremost un-burnt or raw fuel particles present in the exhaust gas, and it considerably depends on fuel ignition system, air-fuel ratio, cetane number of fuel, oxygen content and atomisation characteristics of fuel. The rate of HC emission increased as the engine load increases. The H25 blend shows for compression ratio 14:1–17:1, reduction in hydrocarbon emission by 89.11% at full load condition.

The carbon dioxide (CO₂) emission is frequently known as greenhouse gas (GHG) emission. During combustion process, excess availability of oxygen content results in more CO₂ emission, which is an indication of complete combustion of fuel. In case of biodiesel inherently having less carbon content as compared to conventional diesel fuel

Configuring NO_x in combustion of fuel inside the cylinder mainly due to thermal, prompt and fuel mechanism, at extreme temperature of about 1800K the diatomic nitrogen molecules split into mono atomic nitrogen and combines with the oxygen molecules to form Nitric Oxide (NO) and Nitrogen Dioxide (NO₂) gases.

III. CONCLUSION

1. In the present investigation, it has confirmed that Calophyllum inophyllum oil used as resource to obtain biodiesel. That seeds having more oil content. The extracted oil has more Free Fatty Acids (FFA) about 19%. The Calophyllum inophyllum oil having more viscosity, so it needed two step of esterification.
2. The viscosity of Calophyllum inophyllum oil reduced considerably after transesterification and is comparable to diesel. The experimental result shows that base catalyzed transesterification is a promising area of research for the production of biodiesel in large scale.
3. Biodiesel characteristics like kinematic viscosity, density, flash point, fire point, calorific value and cetane number are comparable to diesel
4. The pure biodiesel (B100) had almost nearer Kinematic viscosity and Calorific value to diesel.
5. Correlations have been established between Kinematic viscosity, flash point, fire point, density, and Calorific value. The Calophyllum inophyllum Methyl Ester (CIME) biodiesel it is observed that it may be used as the alternative fuel in Internal Combustion (IC) engine.
6. Biodiesel has similar combustion characteristics as diesel and biodiesel blend can reduce hydrocarbons, smoke opacity, particulate matters, carbon dioxide and carbon monoxide emissions but NO_x emissions have slightly increased.
7. Competition of edible oil sources as food with fuel makes edible oil not an ideal feedstock for biodiesel production. This shifts the attention to non-edible oil like Jatropha curcas and Calophyllum inophyllum oil which are grown in tropical and subtropical climates country.
8. The present study also concluded that among all types of blend the H25 blend gives the closest values to diesel fuel data and gives better performance and emission characteristics.

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